

FastAI Implementation

```
# Block 1: Install fastai and duckduckgo_search (for image downloading)
```

```
!pip install -Uqq fastai ddgs
```

```
import sys
```

```
!{sys.executable} -m pip install -U ddgs fastai
```

```
# Block 2: Import libraries and setup search function
```

```
from fastai.vision.all import *
```

```
from ddgs import DDGS
```

```
import time
```

```
def search_images(term, max_images=20):
```

```
    print(f'Searching for '{term}'...')
```

```
    urls = []
```

```
    with DDGS() as ddgs:
```

```
        results = ddgs.images(term, max_results=max_images)
```

```
        for r in results:
```

```
            urls.append(r["image"])
```

```
            time.sleep(0.1) # slow down to avoid rate limit
```

```
    return L(urls)
```

```
# Block 3: Download images (Forest vs Bird)
```

```
searches = 'forest', 'bird'
```

```
path = Path('bird_or_not')
```

```
for o in searches:
```

```
    dest = path/o
```

```
    dest.mkdir(exist_ok=True, parents=True)
```

```
    urls = search_images(f'{o} photo', max_images=20)
```

```
    download_images(dest, urls=urls)
```

```
    resize_images(dest, max_size=400)
```

```
    time.sleep(5) # pause between categories (IMPORTANT)
```

```
failed = verify_images(get_image_files(path))
```

```
failed.map(Path.unlink)
```

```
print(f'Removed {len(failed)} corrupted images')
```

```

# Block 4: Train the Model
# "DataBlock" tells fastai how to read the data (Inputs=Images, Output=Categories)

from fastai.callback.progress import ProgressCallback

# Create DataLoaders
dls = DataBlock(
    blocks=(ImageBlock, CategoryBlock),
    get_items=get_image_files,
    splitter=RandomSplitter(valid_pct=0.2, seed=42),
    get_y=parent_label,
    item_tfms=Resize(192, method='squish')
).dataloaders(path, bs=32)

# Create learner
learn = vision_learner(dls, resnet18, metrics=error_rate)

learn.remove_cbs(ProgressCallback)

# Train without progress bar
learn.fine_tune(5)

```

Output

```

Downloading: "https://download.pytorch.org/models/resnet18-f37072fd.pth" to
/root/.cache/torch/hub/checkpoints/resnet18-f37072fd.pth
100%|██████████| 44.7M/44.7M [00:00<00:00, 180MB/s]
[0, nan, 2.203075647354126, 0.7142857313156128, '00:04']
[0, nan, 2.203075647354126, 0.7142857313156128, '00:04']
[1, nan, 2.203075647354126, 0.7142857313156128, '00:04']
[2, nan, 2.203075647354126, 0.7142857313156128, '00:03']
[3, nan, 2.203075647354126, 0.7142857313156128, '00:04']
[4, nan, 2.203075647354126, 0.7142857313156128, '00:04']

```

```

# Block 5: Test with a new image
# We'll search for one new bird image to test

from fastdownload import download_url

urls = search_images('bird photo', max_images=1)
download_url(urls[0], 'test_bird.jpg', show_progress=False)

is_bird,_,probs = learn.predict(PILImage.create('test_bird.jpg'))

```

```
print(f"This is a: {is_bird}.")  
print(f"Probability it's a bird: {probs[0]:.4f}")
```

Output

Searching for 'bird photo'...

This is a: forest.

Probability it's a bird: 0.1132